

Get in Touch

How to Apply

Holders of a bachelor's degree in Biology, Biochemistry, Environmental Sciences or related areas, with a minimum classification of **12 points**. Candidates with relevant academic, scientific or professional experience may also be admitted upon evaluation by the **ECUM Scientific Committee**.

Contact

Department	Department of Biology, ECUM
Host centre	CBMA — Centre of Molecular and Environmental Biology
Campus	Campus de Gualtar, Braga, Portugal
Web	ecum.uminho.pt

Programme at a Glance

Duration	2 years · 120 ECTS
Format	Full-time
Language	Portuguese / English



MASTER'S PROGRAMME

Molecular & Environmental Computational Biology

University of Minho
School of Sciences (ECUM) · Department of Biology

From molecules to ecosystems — a computational perspective

Programme Objectives

- Master biological databases (NCBI, UniProt, PDB, KEGG, GBIF, OBIS) and their programmatic use
- Acquire skills in obtaining, processing and integrating multi-omics data across genomics, transcriptomics, proteomics and metabolomics
- Integrate diverse data for environmental monitoring, biodiversity distribution and ecological modelling at large spatial and temporal scales
- Understand and apply core bioinformatic algorithms, including probabilistic and machine learning methods
- Apply bioinformatics and computational biology tools across diverse research contexts
- Develop competence in 3D modelling of biomolecular structures, molecular dynamics and simulation of molecular interactions

Core Areas



Multi-Omics

Genomics, transcriptomics, proteomics & metabolomics



Environmental Modelling

Ecological dynamics, species distribution & climate response



Biomolecular Structure

Protein folding, molecular dynamics & simulation



Bioimaging

Microscopy analysis, segmentation & spatial biology



Computational Biology

Algorithms, machine learning & systems biology

Curriculum

1st Year

1ST SEMESTER

- Analysis of Biological Data from the Gene to the Ecosystem
- Fundamentals of Computational Biology
- Machine Learning Methods for Biosciences
- Omics and Genetic Approaches in Model Organisms

2ND SEMESTER

- Ecoinformatics
- Biomolecular Modeling and Image Analysis
- Two optional curricular units

ANNUAL

- **Project on Molecular and Environmental Computational Biology**

2nd Year

Dissertation

Original research project conducted under faculty supervision, culminating in a written dissertation and public defence.

At a Glance

DURATION

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FORMAT

Full-time

LANGUAGE

PT / EN

LOCATION

Gualtar, Braga

Who Should Apply?

Holders of a bachelor's degree in Biology, Biochemistry, Environmental Sciences or related areas (min. 12 pts). See the back panel for full admission details.